LECTURE:1 INTRODUCTION TO COMPETITIVE PROGRAMMING

Mondira Chakraborty

Lecturer, CSE Dept.

Notre Dame University Bangladesh

Content

1. What is Competitive Programming?
2. Do we need to learn Competitive Programming?
3. Can competitive programming get you a job?
4. To Start or Not To Start Competitive Programming?
5. Can I get a job at FAANG?
6. What is needed for Competitive Programming?
7. How do I get started with Competitive Programming?
8. What are the different topics to learn in Competitive Programming?
9. What are the best resources to become good at Competitive Programming?
10. Common mistakes to avoid in competitive programming as a beginner
11. Bangla resources for learning competitive programming

What is Competitive Programming?

- Competitive programming is a sport where contestants solve algorithmic problems within a time limit using a programming language of their choice.
- It tests problem-solving skills, knowledge of algorithms, and ability to write efficient code.
- This is popular among students, computer science enthusiasts, and professionals looking to improve their skills.
- It is also used as a means of recruitment by many companies with many hosting their own coding competitions.

Do we need to learn Competitive Programming?

Necessity or need is relative from person to person but here are some points which may motivate you to start competitive programming:

- In recent times competitive programming has gained huge popularity among huge number of organizations along with participants to engage in this sport.
- If you love to have fun solving puzzles with the help of math and coding, competitive programming is the best.
- Most competitive programming contests have prizes. These prizes can be in the form of cash, software, hardware, free services, and more.
- There have even been some sport programmers who had so much success competing in coding contests that they turned it into a career.

Do we need to learn Competitive Programming?

- Many tech companies, like Facebook, Google, and Amazon, use the same type of questions in their technical interviews as those you'll face in coding contests.
- While you may know how to write code, you may not be familiar with the algorithms and data structures that you must use to solve these types of problems.
- There's probably no better way to improve your problem-solving and programming skills than by entering programming contests.
- Competing in programming contests will give you the practice and confidence you need to make it through one of these intensive interviews.

To Start or Not To Start Competitive Programming?

- Well, competitive programming enables and enhances the problem-solving ability of the programmer.
- Organizations search for candidates who have excellent problem-solving capabilities as languages, frameworks, technologies can be learnt at any point of time but the concept or logic behind complex solutions depends upon the thinking capabilities for solving a problem.
- So, although it is not necessary to do competitive programming but it does help in building the problem-solving skills.
- **Caution**: you should only continue it from you love for coding or passionate about it else most of the time people lose interest and leave mid-way without making much progress.

Can I get a job at FAANG (if I start this class)?

What is needed for Competitive Programming?

- Just a knack for programming. Really, all you need is a computer, an internet connection and a knack for programming that's enough to get you started.

- Now, most of the current contests for beginners on various programming sites require some basic knowledge of a programming language to start with.

- Even there are contests for beginners which involve pretty much simple ad-hoc problems which are best for enhancing your confidence.

How do I get started with Competitive Programming?

1. **Learn a Programming Language**

- Start by learning a programming language. Most competitive programmers participate using C/C++.
- Java is the second most popular language for competitive programming. I would suggest you pick either C++ or Java.
- Avoid languages like Python or JavaScript as most of the problems are not tested for those languages and many contests don't even allow those languages.

How do I get started with Competitive Programming?

2. **Practice coding on an online platform**

- Writing code on online judges is very different from how coding is taught in the classroom.
- Additionally, there are certain errors that you would mostly not encounter while coding offline as you would generally not test in a constrained manner while coding offline.
- Warm-up by solving some practice questions which require you to solve logical problems without any prerequisite.

How do I get started with Competitive Programming?

3. **Learn Data Structures and Algorithms(optional)**

- You might want to learn the most common data structures and algorithms before jumping into competitive programming.
- It might help you solve more questions while you participate in contests.
- If you want to start with contests as soon as possible, you may skip this step and learn data structures and algorithms along the way.
- Make sure to at least learn DSA Foundations like how to compute the time and space complexity, recursion, and divide and conquer

How do I get started with Competitive Programming?

4. **Solve practice problems on Toph, CodeForces, UVa etc.**

- Start solving beginner ad-hoc problems (I will help you to find beginner problems).
- You can solve them based on basic maths or simple logic. Solve the first 30 problems from beginner and first 30 problems from easy.
- Initially, it might take some time to solve problems and It is totally fine to stay stuck and ultimately solve after putting in the effort.
- That will make you a better problem-solver. You may even encounter different types of issues.
- Google is your friend.

How do I get started with Competitive Programming?

5. **Start participating in CodeChef Long Challenges**

- At this stage, you should be pretty comfortable with online coding.
- You might not be able to solve many problems, but you will still know the common issues and know how to test and debug.
- In a contest, always start with the easiest problems.
- CodeChef Long Challenge is a great way to learn as you will get 10 days to solve problems.
- It is fine even if you can solve only a single problem.
- The idea is to solve as many problems as possible with time.

How do I get started with Competitive Programming?

5. **Start participating in CodeChef Long Challenges**

- After a contest, go through the editorials of the next 3 problems and try to understand how to solve them.
- After going through the editorial, try to solve those problems in the practice section.
- If that problem requires a concept that you are not aware of (most likely a data structure or algorithm), learn about it and solve a few questions that require that concept.
- Continue doing this as long as possible to keep learning.

How do I get started with Competitive Programming?

6. **Start participating in short contests across websites**

- Shorter contests are the real thing in Competitive Programming.
- You can become a good competitive programmer only by participating in short contests (2 to 2.5 hours).
- After a couple of long contests, you should focus on participating in as many contests as possible across platforms like CodeChef , Codeforces, AtCoder, etc.

How do I get started with Competitive Programming?

6. **Start participating in short contests across websites**

- It is ok even if you are unable to solve a single problem when you start.
- Focus on improving with every contest by going through the editorial and solving the next 2-3 unsolved problems.
- It is a continuous journey of learning. As you start solving more and more problems, the concepts would get more difficult.
- Continue doing this as long as possible to keep getting better.

What are the different topics to learn in Competitive Programming?

1. Introduction to C++
2. Basic DSA a. Arrays b. Linked List c. Stack d. Queue
3. Math for Programming

    a. Fundamentals (Binary Exponentiation, Euclidean Algorithm, etc)

    b. Algorithms related to prime numbers (Sieve of Eratosthenes, etc)

    c. Number Theory (Euler's totient, etc)

    d. Modular Arithmetic (Modulo inverse, Chinese remainder theorem, etc)

    e. Number Systems (Balanced Ternary, Gray code, etc)

    f. Linear Algebra g. Geometry h. Combinatorics

    i. Numerical Methods j. Misc (Fast Fourier transform, Polynomials, etc)

What are the different topics to learn in Competitive Programming?

4. Basic Algorithms a. Searching and Sorting b. Hashing c. 2 pointers d. Backtracking
5. Intermediate Algorithms a. String Manipulation b. Bit Manipulation c. Greedy
6. Intermediate Data Structures a. Trees b. Set c. Map d. Heap
7. Slightly Advanced Algorithms a. Dynamic Programming (DP)
8. Graph DSA

What are the different topics to learn in Competitive Programming?

9. Advanced DSA
   a) Game Theory
   b) Advanced String Algorithms, Tries
   c) Segment Trees, Fenwick Trees
   d) Suffix Tree, Suffix Array
   e) Heavy Light Decomposition
   f) Disjoint Set, Graph Coloring, Network Flow
   g) Sqrt Decomposition

What are the best resources to become good at Competitive Programming?

These are the best resources which you may go through as you get time apart from the contests:

- cp-algorithms - English translation of E-Maxx Algorithms
- Competitive Programming: Steven Halim, Felix Halim
- Competitive Programmer's Handbook by Antti Laaksonen
- Topcoder Competitive Programming Tutorials

Common mistakes to avoid in competitive programming as a beginner

- Focusing too much on the programming language
- All practicing, No learning
- All learning, No practicing
- Only participating in contests
- Not participating in contests
- Solving random questions
- Giving up

Bangla resources for learning competitive programming

1. GitHub - https://github.com/me-shaon/bangla-programming-resources
2. Blog - https://www.shafaetsplanet.com/

References:

1. https://www.codingninjas.com/studio/library/how-to-get-started-with-competitive-programming
2. https://workat.tech/problem-solving/article/competitive-programming-complete-roadmap-from-scratch-sbcrmnxb1sfg
3. https://codeforces.com/blog/entry/111099